

INSTITUTE OF ACTUARIES OF INDIA

EXAMINATIONS

14th September 2021

**Subject CP1 – Actuarial Practice
(Paper A)**

Time allowed: 3 Hours 30 Minutes (09.30 - 13.00 Hours)

Total Marks: 100

- Q. 1)** It has been almost two years of Covid-19 pandemic now however, your company has not written Covid insurance (exclusive short term cover if death happens due to Covid) before and therefore has no data relating to this. It has been suggested that you could use industry-wide data to help you to set the prices.
- i)** Explain the difficulties you might encounter in obtaining and using industry-wide data. (4)
- You have been asked to specify the future data requirements for this class of business so that a suitable management information system can be considered.
- ii)** List the data items for which you would ask in order to help you to develop sophisticated pricing for this business in the future. (4)
- [8]
- Q. 2)** A company currently offers standard Health Insurance cover to all its employees and their dependents. The costs of this cover have risen by 50% in the last five years, and the cover has become unaffordable.
- Discuss ways in which the company can seek to reduce the cost of its cover and likely impact of various approaches (impact quantification is not required). [9]
- Q. 3)** The CEO of your company, who is good at Statistics, has recently heard of the term VaR (Value at risk) and asked you to explain the term.
- i)** Describe “Value at Risk”. (5)
- He has also further requested you to check the VaR for the funding level of your defined benefit scheme which will not fall below a certain level for next three years. You have asked your Actuarial team member to calculate the VaR and prepare a report with the limitations of VaR to enable the CEO understand the risks.
- ii)** Outline the limitations of VaR as a method of measuring risk. (5)
- [10]
- Q. 4)** Tom, Dick and Harry want to open an Antique gift shop. They have conducted an initial appraisal and have concluded that this project is viable and the net present value and payback period are both satisfactory.
- Tom has now suggested that, rather than open one shop, they should open a chain of Antique Gift shops. He states that this would be more profitable and so they should aim for this more ambitious project.
- Discuss this suggestion. [12]
- Q. 5)** **i)** Define the term risk budgeting. (2)
- Due to the unprecedented situation since last two years of Covid-19, an insurance company which predominantly writes Non Linked and Non-participating Policies has been facing a huge investment risk as the yield on G-Sec has reduced drastically and possible future falls in interest rates.

The CEO of the insurance company wants to re-budget their investment risk by writing more Unit Linked Policies to reduce the company's risk as in ULIP policies most of the risk is borne by the policyholders.

- ii)** Outline the risk budgeting process for an investment fund holding a range of asset classes.

(7)

[9]

Q. 6) A large general Insurance Company predominately write Motor and Health business. It has exposure to Equity in the tune of around 35% in term of Book Value in its Investment portfolio. This is very high in comparison to its peers. The Appointed Actuary has highlighted concern of high equity exposure to the management and suggested to reduce it on priority.

- i)** Outline key investment objective of the assets.

(3)

- ii)** List the AA concern of having high equity exposure.

(5)

Regulator has recently raised concern of company solvency. Company solvency has fallen significantly from control level required by regulation in last 2 years.

- iii)** Outline reasons for having low solvency of the company.

(5)

Regulator is concerned of low solvency and to protect policyholder fund has asked company to stop writing new business due to low solvency.

- iv)** Suggest changes in Investment strategy as company entered run-off.

(7)

[20]

Q. 7) **i)** List the cover that can be provided under Cyber Insurance Product.

(5)

- ii)** Outline the risk factors to consider in pricing cyber risk.

(7)

A large general insurance company is looking to start writing insurance for cyber risk.

iii)

- a)** Outline the opportunities of entering cyber risk business.

(2)

- b)** Outline the pricing challenges and risks the company faces moving into this new business line.

(4)

- c)** Suggest how the company might manage potential risks of entering this business.

(4)

[22]

Q. 8) Regulator of a country is proposing an Industry level body which will collect both policy and claims data of Health from all general Insurance companies.

Organization will publish report yearly highlighting age band wise average claim paid amount and disease wise claim summary.

- i)** Outline the benefits of using industry-wide collection scheme. (4)
 - ii)** Outline the limitation of such scheme in meeting its objective. (6)
- [10]**
